JOHNS HOPKINS
WHITING SCHOOL
of ENGINEERING

Electrical & Computer Engineering 520.445/645
Audio Signal Processing
Fall, 2026

Project 3
Head-Related Transfer Functions

Deadline: 10/27/2026 at 11:59pm

Project Overview
The goal of this project is to explore the fundamental concepts and practical applications of Head-Related Transfer Functions (HRTF) and Head-Related Impulse Responses (HRIR). You will work with a dataset of HRTF files recorded from multiple subjects and use them to create and analyze spatial audio effects.

Dataset
You are provided with two sets of data: HRTFs and audio samples.

The HRTF data is taken from the following article:
Doma, Shaimaa, Natálie Brozova, and Janina Fels. "Examining the interrelation behavior of distance metrics for head-related transfer function evaluation: a case study", Acta Acustica, Volume 7, 2023. DOI: https://doi.org/10.1051/aacus/2023028
The supplementary material in the article provides the data used in this project. No. RWTH-2023-05668. Lehrstuhl für Hörtechnik und Akustik, 2023.

The audio samples are sourced from freesound.org and various music productions, resampled to 48kHz in mono.

All files are included in that attached zip file organized as follows:
Data -> Audio -> music1.wav
Data -> HRTFs -> id01_measured.sofa

Project Tasks

1. HRTF overview
Do your own reading/research and write a summary (maximum ½ page, single spaced) explaining:
- what HRTFs are
- How they are measured and represented
- Their main application in spatial and 3D audio systems

No AI-tools are allowed to write your summary.

2. Visualizing HRTF data
Load and Understand HRTF Files:
a. Explore the structure of the files by checking the indices of Elevation and Azimuth. Load the HRIR data for a specific direction and a given ear (read the reference article)
b. Convert between HRTF and HRIR using FFT and IFFT
    - Note: The toolkit should load the HRIR directly from the SOFA file.

c. Apply an HRIR filter to a given audio clip
d. Visualize the HRTF data from different subjects:
    - Azimuth HeatMap: Keep Elevation constant, plot a magnitude heatmap with the x-axis as Azimuth (in degrees) and y-axis as frequency (in Hz)
    - Elevation HeatMap: Keep Azimuth constant, plot a magnitude heatmap with the x-axis as Elevation (in degrees) and y-axis as frequency (in Hz)
    - Note: If your heatmap appears unclear, consider switching to a log scale for better visualization
e. Write a short analysis comparing the similarities and differences between these HRTF files across subjects. (Discuss at least 3 aspects, such as common patterns between magnitude and azimuth/elevation, inter-subject differences, frequency-dependent differences and any other elements that stand out to you)
f. Mean HRTF:
   Compute the mean magnitude across all subjects for the azimuth and elevation Heatmaps explored above and visualize them.
   Calculate the MSE (mean square error) between each subject's HRTF and the mean HRTF along azimuth and elevation. Identify the subject closest to the mean and the subject farthest from the mean HRTF along each direction.

3. Surround sound using HRTFs

Build a tool that:

- Takes an HRTF configuration file (for a given subject), an audio sample, a starting azimuth angle and an azimuth rotation speed (angular velocity in degrees per second), basically simulating an sound source moving in space.
- Produces a spatial (surround sound) effect by applying different HRTF configurations across time frames.
    - Note: we keep elevation fixed and focus on rotating azimuth.
- When synthesizing audio, make sure to use overlap-add (OLA) windowing or similar strategies to ensure smooth transitions and avoid clipping artifacts.

Refer to the provided example file for guidance on achieving continuous and realistic spatial movement.

Your tool can be a simple function with informative inputs or can be a simple graphical interface. It is up to you!

4. Experiments and listening evaluation

Apply your surround sound tool to the audio samples provided. Test at least 5 subjects, including:

- The subject closest to the mean HRTF
- The subject farthest from the mean HRTF
- Three randomly selected subjects

a. Listening Analysis

Compare your spatial audio outputs with the earlier visualizations and Mean HRTF analysis. Discuss at least 3 aspects, such as:

a. How the perceived direction matches your azimuth trajectory
b. Variations in loudness or frequency balance with direction
c. How the sound differs for subjects farthest from the mean

b. Preferred HRTF

Identify your preferred HRTF and explain your choice using at least 3 criteria, such as:

a. Sound quality
b. Positional accuracy
c. Smoothness during azimuth changes

Deliverables (on canvas):

1. A report (5 pages)

- Section1: Summary of HRTF concept and understanding
- Section 2: visualization of HRTF data, analysis of inter-subject differences, mean and MSE
- Section 3: description of your surround sound tool
- Section 4: Listening experiment results and discussion of preferred HRTF configuration

2. Generated spatial audio files using your surround sound tool
3. link to your code (e.g. google collab) or submit your code directly. You are free to use MATLAB or python. Make sure your code is working properly and includes all functions necessary to properly test your system. If your code crashes for any reason, points will be deducted.
4. If your files are big, you can upload them to one drive and include a link in your submission. Make sure the files are accessible by the instructor and TAs.

Grading Rubrics

Components		**Points**
Summary	HRTF summary	10
Visualization	Visualization code and plots	10
	Visualization analysis	5
Mean HRTF	Code to compute mean HRTF	7.5
	Code to compute distances	7.5
Surround sound tool	Code works for applying HRIR filters	10
	Code works for creating surround sound	15
	Reasonable sound quality and smooth transition	10
Listening Analysis	HRTFs comparison and Analysis	10
	Discussion of favorite HRTF Configuration and why	10
Total		**100**